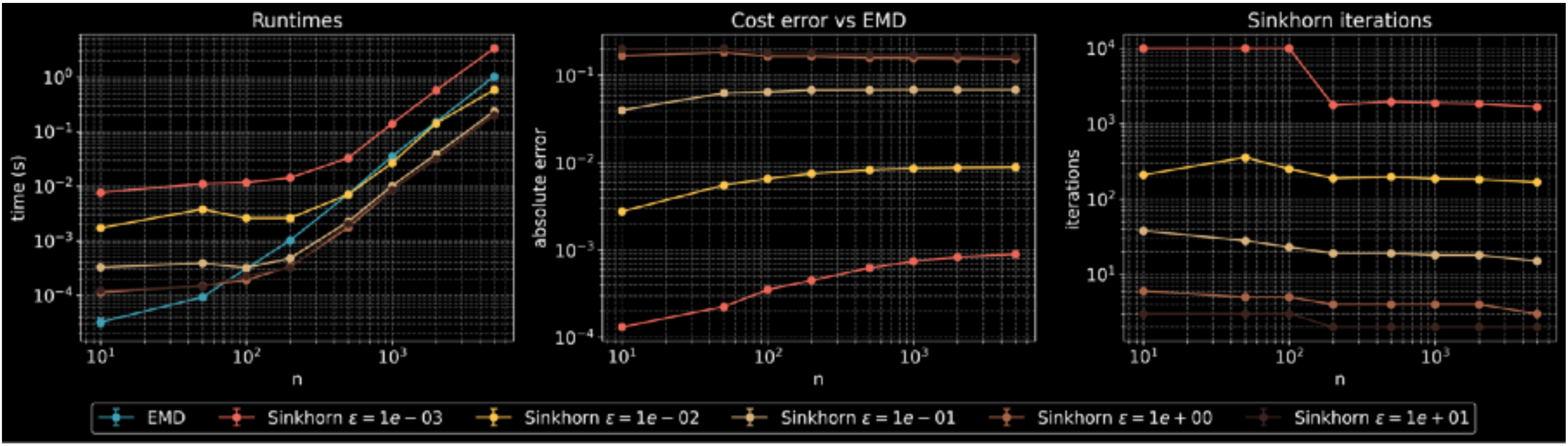

Runtimes
time (s)
n
Cost error vs EMD
absolute error
n
Sinkhorn iterations
iterations
n
EMD
Sinkhorn $\varepsilon = 1e-03$
Sinkhorn $\varepsilon = 1e-02$
Sinkhorn $\varepsilon = 1e-01$
Sinkhorn $\varepsilon = 1e+00$
Sinkhorn $\varepsilon = 1e+01$

Theorem [Sinkhorn Global Convergence]: One has $(\mathbf{u}^{(\ell)}, \mathbf{v}^{(\ell)}) \rightarrow (\mathbf{u}^*, \mathbf{v}^*)$ and

$$d_{\mathscr{H}}(\mathbf{u}^{(\ell)}, \mathbf{u}^*) = O(\lambda(\mathbf{K})^{2\ell})$$

$$d_{\mathscr{H}}(\mathbf{v}^{(\ell)}, \mathbf{v}^*) = O(\lambda(\mathbf{K})^{2\ell})$$

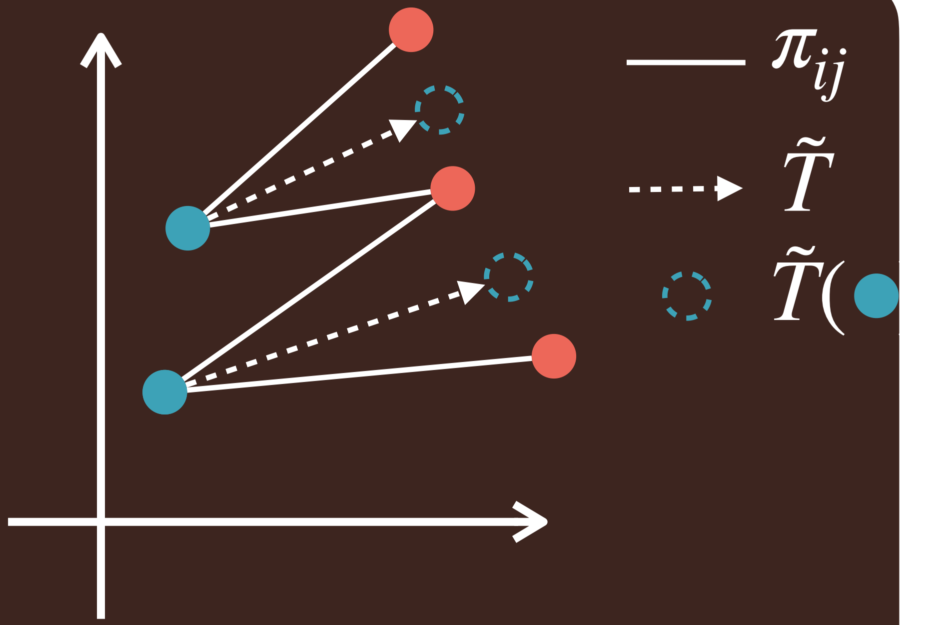
I.e., the iterates converge geometrically to optima.

The effects of ε in EOT

- small $\varepsilon \rightarrow$ accurate but unstable/slow.
- large $\varepsilon \rightarrow$ stable/fast but approximate.

The Barycentric Projection

- $\tilde{T} : \mathbf{x}_i \mapsto \frac{1}{a_i} \sum_j \pi_{ij} \mathbf{y}_j$
- $\tilde{T}(\mathbf{x}) = \mathbb{E}_\pi[Y \mid X = \mathbf{x}]$
- ‘approximate Monge map’
- Can be extended out of sample for entropic OT



Entropic OT Dual

$$\max_{\mathbf{f},\mathbf{g}} \langle \mathbf{f}, \mathbf{a} \rangle + \langle \mathbf{g}, \mathbf{b} \rangle - \varepsilon \sum_{i,j} \exp\left(\frac{f_i + g_j - C_{ij}}{\varepsilon} \right)$$

Log-Domain Sinkhorn

$$\mathbf{f}_i = \varepsilon \log \mathbf{a}_i + \min\nolimits_\varepsilon(\mathbf{C}_{i,:} - \mathbf{g})$$

$$\mathbf{g}_j = \varepsilon \log \mathbf{b}_j + \min\nolimits_\varepsilon(\mathbf{C}_{:,j} - \mathbf{f})$$

Last Time

The effects of ε in EOT

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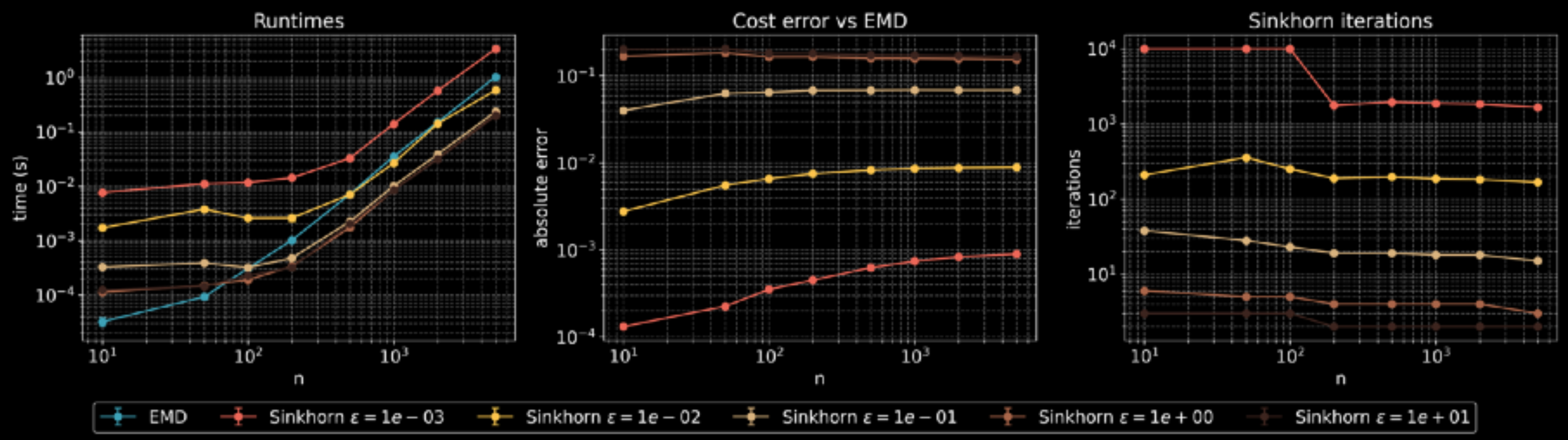

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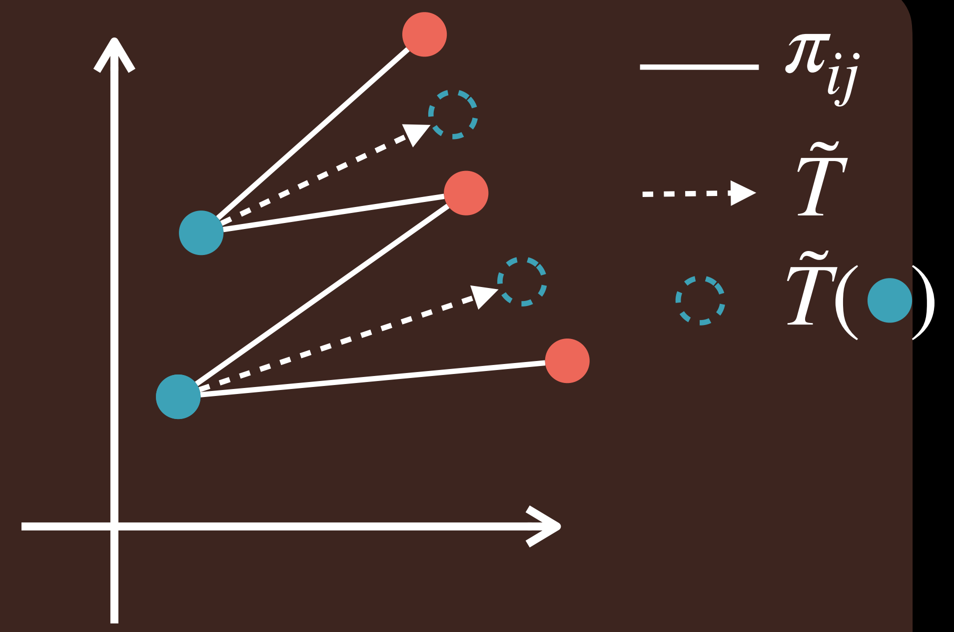

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Today

- **Sinkhorn Divergences** deep dive: motivation, properties, relation to MMD
- **Wasserstein Barycenters:** geometry-aware averages of distributions